

Exercises 1-1 Answer

25.

(5) $f(x) = \sin(x^2)$ is **NOT** periodic.

(6) $y = \sin^4 x + \cos^4 x$ **IS** periodic.

$$\begin{aligned} y &= \sin^4 x + \cos^4 x \\ &= (\sin^2 x + \cos^2 x)^2 - 2 \sin^2 x \cos^2 x \\ &= 1 - \frac{1}{2} \sin^2 2x \\ &= \frac{3}{4} + \frac{1}{4} \sin 4x. \\ T &= \frac{\pi}{2}. \end{aligned}$$

26.

(2) **IS** elementary function.

$$y = \begin{cases} -x - x^3, & x < 0, \\ x + x^3, & x \geq 0. \end{cases} = \sqrt{(x + x^3)^2}.$$

(3) **NOT** elementary function.

$$y = \begin{cases} x^2, & x < 1, \\ 0, & x \geq 1. \end{cases}$$

(4) **NOT** elementary function.

$$y = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R}/\mathbb{Q}. \end{cases}$$

Exercises 1-2 Answer

2.

(2)

$$\lim_{n \rightarrow \infty} \frac{\sqrt[3]{n^2} \sin n!}{(n+1)^2} = 0.$$

pf. let $N = \varepsilon^{-\frac{3}{4}}$,

$$\begin{aligned} \forall \varepsilon > 0, n > N = \varepsilon^{-\frac{3}{4}}, \\ \varepsilon > \frac{1}{n^{\frac{4}{3}}} > \frac{1}{n^{\frac{4}{3}} + 2n^{\frac{1}{2}} + n^{-\frac{2}{3}}} = \frac{\sqrt[3]{n^2}}{(n+1)^2} \geq \left| \frac{\sqrt[3]{n^2} \sin n!}{(n+1)^2} - 0 \right|. \end{aligned}$$

By definition of a limit, the original limit is 0. ■

(3)

$$\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0.$$

pf. let $N = \frac{1}{4\varepsilon^2}$,

$$\forall \varepsilon > 0, n > N = \frac{1}{4\varepsilon^2},$$

$$|(\sqrt{n+1} - \sqrt{n}) - 0| = \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{2\sqrt{n}} < \varepsilon.$$

By definition of a limit, the original limit is 0. ■

(4)

$$\lim_{n \rightarrow \infty} \frac{n}{a^n} = 0 \quad (a > 1).$$

pf. let $a = h + 1$,

$$(h+1)^n = 1 + nh + \frac{n(n-1)}{2}h^2 + \dots,$$

$$n > \frac{2}{\varepsilon h^2} + 1$$

let $N = \frac{2}{\varepsilon(a-1)^2} + 1$.

$$\forall \varepsilon > 0, n > N = \frac{2}{\varepsilon(a-1)^2} + 1,$$

$$\left| \frac{n}{a^n} - 0 \right| = \frac{n}{a^n} = \frac{n}{((a-1)+1)^n} < \frac{2}{(n-1)h^2} < \varepsilon.$$

By definition of a limit, the original limit is 0. ■

3. **pf.**

$$\because \lim_{n \rightarrow \infty} a_n = a,$$

$$\therefore \forall \varepsilon > 0, \exists N, s.t. n > N, |a_n - a| < \varepsilon.$$

$$\because n + m > N + m > N,$$

$$\therefore \forall \varepsilon > 0, \exists N, s.t. n > N, |a_{n+m} - a| < \varepsilon.$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_{n+m} = a. \blacksquare$$

4. **pf.**

$$1^\circ \because \lim_{n \rightarrow \infty} a_n = a,$$

$$\therefore \forall \varepsilon > 0, \exists N s.t. n > N, |a_n - a| < \varepsilon.$$

$$\because ||a_n| - |a|| \leq |a_n - a| < \varepsilon,$$

By definition, $\lim_{n \rightarrow \infty} |a_n| = |a|$.

$$\begin{aligned}
2^\circ \quad a_n &= (-1)^n, \\
\lim_{n \rightarrow \infty} |(-1)^n| &= 1, \\
\lim_{n \rightarrow \infty} (-1)^n &= DNE.
\end{aligned}$$

$$3^\circ \quad \lim_{n \rightarrow \infty} |a_n| = 0, \forall \varepsilon > 0, \exists N \text{ s.t. } n > N,$$

$$||a| - 0| < \varepsilon \Leftrightarrow |a_n| < \varepsilon \Leftrightarrow |a_n - 0| < \varepsilon$$

By definition, $\lim_{n \rightarrow \infty} a_n = 0$. ■

7.

(3)

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \frac{1 + a + a^2 + \cdots + a^n}{1 + b + b^2 + \cdots + b^n} \\
&= \lim_{n \rightarrow \infty} \frac{1 - a^{n+1}}{1 - a} \frac{1 - b}{1 - b^{n+1}} \\
&= \frac{1(1 - b)}{(1 - a)1} \\
&= \frac{1 - b}{1 - a}.
\end{aligned}$$

(4)

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \left(\frac{1}{n^2} + \frac{2}{n^2} + \cdots + \frac{n-1}{n^2} \right) \\
&= \lim_{n \rightarrow \infty} \frac{n(n-1)}{2n^2} \\
&= \lim_{n \rightarrow \infty} \left(\frac{1}{2} - \frac{1}{2n} \right) \\
&= \frac{1}{2}.
\end{aligned}$$

(5)

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \left(\frac{1^2}{n^3} + \frac{2^2}{n^3} + \cdots + \frac{(n-1)^2}{n^3} \right) \\
&= \lim_{n \rightarrow \infty} \frac{(n-1)n(2n-1)}{6n^3} \\
&= \lim_{n \rightarrow \infty} \left(\frac{1}{3} - \frac{1}{2n} + \frac{1}{6n^2} \right) \\
&= \frac{1}{3}.
\end{aligned}$$

(6)

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left(\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \cdots + \frac{1}{n(n+1)} \right) \\ &= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots + \frac{1}{n} - \frac{1}{n+1} \right) \\ &= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right) \\ &= 1. \end{aligned}$$

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25. (5) $f(x) = \sin x^2$, NO

(6) $y = \sin^4 x + \cos^4 x$

$$= (\sin^2 x + \cos^2 x)^2 - 2 \sin^2 x \cos^2 x$$

$$= 1 - \frac{1}{2} \sin^2 2x$$

$$= \frac{3}{4} + \frac{1}{4} (1 - 2 \sin^2 2x)$$

$$= \frac{3}{4} + \frac{1}{4} \sin 4x$$

$$T = \frac{2\pi}{4} = \frac{\pi}{2}$$

26. (3) $y = \begin{cases} x^2, & x < 1, \\ 0, & x \geq 1 \end{cases}$ NO

(2) $y = \begin{cases} -x-x^3, & x < 0 \\ x+x^3, & x \geq 0 \end{cases} = \sqrt{(x+x^3)^2}$

YES

(4) $y = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \in \mathbb{R}/\mathbb{Q} \end{cases}$

$$2(2) \lim_{n \rightarrow \infty} \frac{\sqrt[3]{n^2} \sin n!}{(n+1)^2} = 0$$

pf. $\forall \varepsilon > 0$

$$\left| \frac{\sqrt[3]{n^2} \sin n!}{(n+1)^2} \right| \leq \varepsilon$$

$$\leq \frac{\sqrt[3]{n^2}}{n^2 + 2n + 1} = \frac{1}{n^{\frac{4}{3}} + 2n^{\frac{1}{3}} + n^{-\frac{2}{3}}} \leq \frac{1}{n^{\frac{4}{3}}} < \varepsilon$$

$$n^{\frac{4}{3}} > \frac{1}{\varepsilon}$$

$$n > \varepsilon^{-\frac{3}{4}}$$

$$\rightarrow \forall \varepsilon > 0, \text{ when } n > N = \varepsilon^{-\frac{3}{4}}$$

$$\begin{aligned} \varepsilon &> \frac{1}{n^{\frac{4}{3}}} > \frac{1}{n^{\frac{4}{3}} + 2n^{\frac{1}{3}} + n^{-\frac{2}{3}}} \\ &= \frac{\sqrt[3]{n^2}}{(n+1)^2} \geq \left| \frac{\sqrt[3]{n^2} \sin n!}{(n+1)^2} \right| \end{aligned}$$

$$(3) \lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0$$

$\forall \varepsilon > 0$

$$|\sqrt{n+1} - \sqrt{n}| < \varepsilon$$

$$\Leftrightarrow \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{2\sqrt{n}} < \varepsilon$$

$$\rightarrow \forall \varepsilon > 0, \text{ when } n > N = \frac{1}{4\varepsilon^2}$$

$$|\sqrt{n+1} - \sqrt{n}| = \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{2\sqrt{n}} < \varepsilon$$

$$(4) \lim_{n \rightarrow \infty} \frac{n}{a^n} = 0 \quad (a > 1)$$

pf. $\left| \frac{n}{a^n} \right| = \frac{n}{a^n} < \varepsilon$

let $a = h + 1$

$$(h+1)^n = 1 + nh + \frac{n(n-1)}{2}h^2 + \dots$$

$$\frac{n}{a^n} < \frac{1}{\frac{n-1}{2}h^2} = \frac{2}{(n-1)h^2} < \varepsilon$$

$$n > \frac{2}{\varepsilon h^2} + 1$$

$$\rightarrow \forall \varepsilon > 0, \text{ when } n > N = \frac{2}{\varepsilon(h+1)^2} + 1$$

$$\left| \frac{n}{a^n} \right| = \frac{n}{a^n} = \frac{n}{[(h+1)+1]^n}$$

$$< \frac{2}{(n-1)h^2}$$

$$< \varepsilon$$

3 pf. $\therefore \lim_{n \rightarrow \infty} a_n = a$

$\therefore \forall \varepsilon > 0, \exists N$ s.t. when $n > N, |a_n - a| < \varepsilon$

$\therefore n+m > N+m > N$

$\therefore \forall \varepsilon > 0$, when $n+m > N+m > N, |a_{n+m} - a| < \varepsilon$

$\Rightarrow \lim_{n \rightarrow \infty} a_{n+m} = a \quad \square$

4 pf. 1° $\therefore \lim_{n \rightarrow \infty} a_n = a$

$\therefore \forall \varepsilon > 0, \exists N$ s.t. when $n > N, |a_n - a| < \varepsilon$

$||a_n| - |a|| \leq |a_n - a| < \varepsilon$

$\therefore$ by definition $\lim_{n \rightarrow \infty} |a_n| = |a|$

2° if $a_n = (-1)^n$

$\lim_{n \rightarrow \infty} |(-1)^n| = 1$

$\lim_{n \rightarrow \infty} (-1)^n = \text{DNE}$

3° $\lim_{n \rightarrow \infty} |a_n| = 0$

$\forall \varepsilon > 0, \exists N$ s.t. when $n > N$

$||a_n| - 0| < \varepsilon \Leftrightarrow |a_n| < \varepsilon \Leftrightarrow |a_n - 0| < \varepsilon$

by definition $\lim_{n \rightarrow \infty} a_n = 0$

$$\begin{aligned}
 7 \quad (3) \quad & \lim_{n \rightarrow \infty} \frac{1 + a + a^2 + \dots + a^n}{1 + b + b^2 + \dots + b^n} \\
 &= \lim_{n \rightarrow \infty} \frac{1 - a^{n+1}}{1 - a} \cdot \frac{1 - b}{1 - b^{n+1}} \\
 &= \frac{1 \cdot (1 - b)}{(1 - a) \cdot 1} \\
 &= \frac{1 - b}{1 - a}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & \lim_{n \rightarrow \infty} \left[\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} \right] \\
 &= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{n} - \frac{1}{n+1} \right) \\
 &= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right) \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 (4) \quad & \lim_{n \rightarrow \infty} \left(\frac{1}{n^2} + \frac{2}{n^2} + \dots + \frac{n-1}{n^2} \right) \\
 &= \lim_{n \rightarrow \infty} \frac{n(n-1)}{2n^2} \\
 &= \lim_{n \rightarrow \infty} \left(\frac{1}{2} - \frac{1}{2n} \right) \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 (5) \quad & \lim_{n \rightarrow \infty} \left[\frac{1^2}{n^3} + \frac{2^2}{n^3} + \dots + \frac{(n-1)^2}{n^3} \right] \\
 &= \lim_{n \rightarrow \infty} \frac{(n-1)n(2n-1)}{6n^3} \\
 &= \lim_{n \rightarrow \infty} \frac{2n^3 - 3n^2 + n}{6n^3} \\
 &= \lim_{n \rightarrow \infty} \left(\frac{1}{3} - \frac{1}{2n} + \frac{1}{6n^2} \right) \\
 &= \frac{1}{3} - \lim_{n \rightarrow \infty} \left(\frac{1}{6n^2} - \frac{1}{2n} \right) \\
 &= \frac{1}{3}
 \end{aligned}$$